

★ Population Matrix System

Model discrete population interactions via system of linear equations
Convert to matrix form: $\vec{v}_{n+1} = A\vec{v}_n$
 A = Transition matrix containing interaction rates
Row equations represent next year's population based on current year
⚠ Ensure variables align exactly to matrix columns when extracting constants

★ Eigenvalues & Stability

Find eigenvalues by solving: $\det(A - \lambda I) = 0$
Find associated eigenvectors by solving: $(A - \lambda I)\vec{x} = 0$
Dominant Eigenvalue (λ_{max}) dictates long-term survival
 $\lambda_{max} > 1 \implies$ Continuous growth
 $\lambda_{max} < 1 \implies$ Population eventually decays to zero
 $\lambda_{max} = 1 \implies$ Stable steady state

Critical Rate of Predation

Condition where a population is perfectly stable (neither growing nor decaying)
Set the dominant eigenvalue $\lambda = 1$
Substitute $\lambda = 1$ back into characteristic equation $\det(A - \lambda I) = 0$
Solve for the unknown predation constant c
⚠ Always specify which species' predation rate is being altered

★ TS Decomposition Models

Components: Trend (T_t), Seasonal (S_t), Cyclical (C_t), Irregular (I_t)
Additive Model: $Y_t = T_t + S_t + C_t + I_t$
Use additive when seasonal amplitude is relatively constant over time
Multiplicative Model: $Y_t = T_t \times S_t \times C_t \times I_t$
Use multiplicative when seasonal amplitude scales with the trend level

TS Component Extraction

Detrending: Subtracting trend component from data ($Y_t - T_t$)
Seasonal Index: Average seasonal effect for a specific recurring period
Seasonally Adjusted: Series with the seasonal component removed
Calculated via moving averages (e.g., 2×4 point MA for quarterly data)

★ Stationarity in TS

Stationary Series: Mean, variance, and autocorrelation remain constant over time
Trend Stationary: Series has a deterministic mean trend
Difference Stationary: Series has a stochastic trend (random walk behavior)
⚠ Many ARIMA models require strict stationarity before fitting

Rendering TS Stationary

For Trend Stationary: Fit a regression trend line and subtract it from the data
For Difference Stationary: Take the difference between consecutive observations
Formula: $\Delta y_t = y_t - y_{t-1}$
If variance is non-constant, apply a logarithmic transformation first

★ ACF & PACF Basics

ACF (Auto-Correlation Function): Correlation of a time series with its own past values (lags)
PACF (Partial Auto-Correlation): Correlation of a series with a lag, removing effects of intermediate lags
Used primarily to determine parameters p and q for ARMA models

★ Identifying AR(p) & MA(q)

AR(p) Model: PACF cuts off sharply after lag p ; ACF tails off gradually
MA(q) Model: ACF cuts off sharply after lag q ; PACF tails off gradually
ARMA(p,q): Both ACF and PACF tail off gradually
⚠ A cut off means bars drop to zero immediately after a specific lag

Information Criteria

Metrics used to evaluate and compare ARIMA model fits
AIC (Akaike) & BIC (Bayesian) penalize the inclusion of extra parameters
Formula structure: Error Term + Penalty for Complexity
Goal: Select the model with the lowest AIC/BIC value
Provides quantitative method when visual inspection is ambiguous

★ SARIMA Model Signatures

Model format: $ARIMA(p, d, q)(P, D, Q)_m$
 m = Seasonal period (e.g., 12 for monthly, 4 for quarterly)
Look for significant spikes at multiples of m (lags 12, 24, 36)
Significant ACF spike at lag $m \implies$ Seasonal MA(1) component
Significant PACF spike at lag $m \implies$ Seasonal AR(1) component

Ex Post vs Ex Ante

Ex Post: Forecasts made using historical data where true values are already known
Used for validation, backtesting, and computing error metrics (Gold Standard)
Ex Ante: Forecasts predicting true future values where actuals are completely unknown
Relies entirely on estimated coefficients holding true in the future

Forecast Types

Qualitative: Subjective, based on expert opinions, surveys, and intuition
Used when historical data is unavailable or irrelevant (e.g., new products)
Quantitative: Objective, mathematical models applied to historical numerical data
Assumes past patterns will continue into the future

Naïve Forecasting Model

Formula: $\hat{y}_{t+1} = y_t$
Next period forecast equals the last observed value
Highly effective for random walks
Used universally as a baseline benchmark to evaluate more complex models
⚠ If a complex model cannot beat Naïve, it is useless

★ Exponential Smoothing

Formula: $\hat{y}_{t+1} = \alpha y_t + (1 - \alpha)\hat{y}_t$
 α = Smoothing constant where $0 < \alpha \leq 1$
EWMA stands for Exponentially Weighted Moving Average
Higher α places more weight on recent observations (reacts faster)
Lower α places more weight on historical history (smoother line)
⚠ Reduces exactly to the Naïve model when $\alpha = 1$

★ Complex Systems

System composed of many interacting parts (agents)
Global macro-level behavior cannot be deduced solely from individual micro-level parts
Exhibits highly non-linear dynamics
Extremely sensitive to initial conditions and environmental changes

Emergence vs Self-Org

Emergence: Macro-level properties arising from local micro-level interactions (e.g., flocking)

Self-Organization: Spontaneous formation of ordered patterns without central control

⚠ Self-organization is the process; Emergence is the result
Self-organization can exist without emergence (e.g., crystallization)

System Modelling Steps

1. Define the purpose and problem boundary
2. Identify the agents and the environment
3. Define interaction rules (agent-agent and agent-environment)
4. Implement as a computational simulation
5. Validate model output against real-world macro observations

★ Cellular Automata (CA)

Grid-based models operating in discrete space and discrete time
Cells are stationary and hold discrete states (e.g., 0=empty, 1=tree)
State transitions depend entirely on local neighborhood rules
Excellent for modeling spatial dynamics and diffusion (e.g., forest fires)

CA Neighborhoods

Von Neumann Neighborhood: The 4 orthogonally adjacent cells (North, South, East, West)

Moore Neighborhood: The 8 surrounding cells (includes diagonals)
Sum of states in neighborhood (S) dictates central cell's next state
Space can have no boundary (infinite) or Toroidal boundaries (wrap-around)

CA Fire Dimensions

1-D Space: Single line of cells. Fire halts permanently upon hitting one empty gap. Easily contained.

2-D Space: Grid of cells. Fire has multiple paths.

3-D Space: Cube of cells. Exponentially more paths for propagation.

⚠ 3-D increases emergent complexity, making containment extremely difficult

★ Agent-Based Models (ABM)

Simulates autonomous, mobile, heterogeneous agents
Agents have internal memory, distinct states, and individual goals
Operate in continuous or discrete space environments
Agents adapt behavior based on local interactions
Contrasts with CA because agents move and are not uniform

ABM Flocking (Boids)

Three core rules generate emergent flocking behavior:

Separation: Steer to avoid collision/crowding with local flockmates

Alignment: Steer to match the average velocity and heading of local flockmates

Cohesion: Steer towards the average position (center of mass) of local flockmates

★ Dynamical Attractors

Attractor: Set of states toward which a system naturally evolves over time

Point Attractor: Settles to a single constant equilibrium value

Limit Cycle: Settles into a closed, repeating periodic loop

Strange Attractor: Fractal structure exhibiting chaotic, non-repeating trajectories sensitive to initial conditions

Limit Cycle Predator-Prey

Models Lotka-Volterra interactions

Creates a perpetual closed-loop trajectory in phase space

Cycle: Prey peaks $\Rightarrow$ Predators spike $\Rightarrow$ Prey overhunted $\Rightarrow$

Predators starve $\Rightarrow$ Prey recovers

Loop repeats indefinitely

★ Logistic Map Equation

Formula: $x_{n+1} = rx_n(1 - x_n)$

Discrete-time demographic model

x_n = Current population ratio ($0 \leq x \leq 1$)

r = Combined rate of reproduction and starvation

Demonstrates how simple deterministic non-linear equations yield chaotic behavior

Bifurcation Diagrams

Plots long-term stable values of x against parameter r

Critical Points: Values of r where period-doubling occurs

Chaotic Region: Dense bands indicating unpredictability (usually $r > 3.57$)

Windows of Stability: Clear vertical strips within chaos where predictable periodicity returns

Network Communities

Identifying dense clusters within a larger graph

Nodes in the same community are highly interconnected

Edges between different communities are sparse

Used to find social circles, biological modules, or echo chambers

★ Modularity & Algorithms

Modularity (Q): Metric scoring community division quality. Higher Q is better.

Louvain Method: Bottom-up algorithm; iteratively merges nodes to maximize Modularity greedily

Girvan-Newman: Top-down algorithm; iteratively removes edges with highest betweenness centrality

⚠ If every node is isolated, Q is zero or negative

★ Markov Assumptions

Stochastic process adhering to the Memoryless Property

$P(X_{t+1}|X_t, X_{t-1} \dots) = P(X_{t+1}|X_t)$

Future state depends solely on the present state

Assumes finite number of discrete states

Assumes time-homogeneous (constant) transition probabilities

Markov Matrices

Transition Matrix (P): Square matrix where P_{ij} is probability of moving from i to j

Every row MUST sum to 1

State Vector (v_t): Represents probability distribution of current states

To find probabilities after n steps: multiply initial state by P^n ($v_n = v_0 P^n$)

★ MCMC Basics

Markov Chain Monte Carlo: Method to draw samples from complex probability distributions

Constructs a Markov Chain whose equilibrium distribution matches the target

Used in Bayesian inference: Posterior $\propto$ Prior $\times$ Likelihood

Estimates Posterior without computing difficult normalizing constants

MCMC Samplers

Role of Sampler: Proposes random walk through state space and decides to accept or reject

Metropolis-Hastings: Accepts new state based on probability ratio
If more probable, accept. If less, accept with some probability.

Advantage: Explores high-probability regions efficiently while sampling entire space